

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2023/2024

SEPTEMBER EXAMINATION

UCCD2063 ARTIFICIAL INTELLIGENCE TECHNIQUES

MONDAY, 2 OCTOBER 2023

TIME : 2.00 PM – 4.00 PM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING

Instructions to Candidates :

This question paper consists of FIVE (5) questions.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**. Each question carries 25 marks.

Should a candidate answers more than ONE (1) question in section B, marks will only be awarded for the FIRST ONE (1) question in that section in the order the candidate submits the answers.

Candidates are allowed to use a calculator.

Answer questions only in the answer booklet provided.

UCCD2063 ARTIFICIAL INTELLIGENCE TECHNIQUES**Section A (Answer All Questions)**

- Q1. (a) Consider a weather model with three random variables: Rain (R), Wind (W), and Temperature (T). The joint probabilities for different combinations of these variables are as follows:

$$P(R = \text{True}, W = \text{Strong}, T = \text{Cold}) = 0.15$$

$$P(R = \text{True}, W = \text{Weak}, T = \text{Cold}) = 0.05$$

$$P(R = \text{True}, W = \text{Strong}, T = \text{Hot}) = 0.2$$

$$P(R = \text{True}, W = \text{Weak}, T = \text{Hot}) = 0.1$$

$$P(R = \text{False}, W = \text{Strong}, T = \text{Cold}) = 0.05$$

$$P(R = \text{False}, W = \text{Weak}, T = \text{Cold}) = 0.1$$

$$P(R = \text{False}, W = \text{Strong}, T = \text{Hot}) = 0.15$$

$$P(R = \text{False}, W = \text{Weak}, T = \text{Hot}) = 0.2$$

Using probabilistic inference, answer the following questions. Show your calculation steps in the answers.

- (i) What is the probability of strong wind? (2 marks)
- (ii) What is the probability of strong wind and cold? (2 marks)
- (iii) What is the probability of rain given that the wind is strong and the temperature is cold? (4 marks)
- (iv) What is the probability of rain given that the wind is strong? (4 marks)

- (b) Figure 1.1 shows a Bayesian Network:

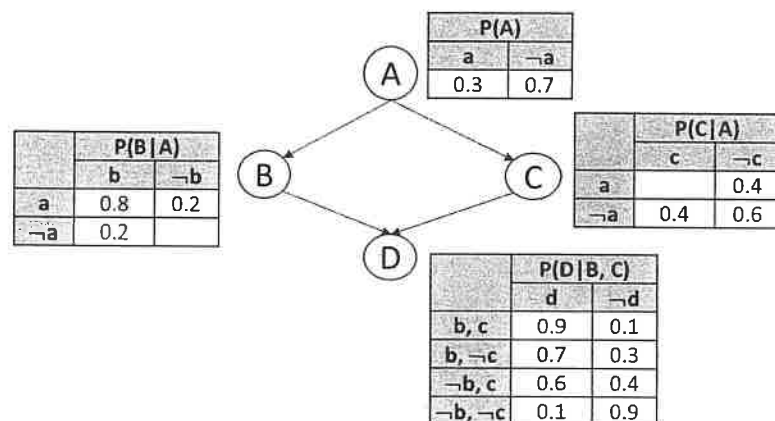


Figure 1.1. A Bayesian Network

- (i) Identify the conditional independent relationship between A and D in the network. (2 marks)
- (ii) Compute the probability for $P(\neg a, \neg b, \neg c, \neg d)$. (3 marks)

UCCD2063 ARTIFICIAL INTELLIGENCE TECHNIQUES**Q1. (b) (Continued)**

- (iii) Compute the probability for $P(b, c, d)$. (4 marks)
- (iv) Compute the probability for $P(b, c \mid a)$. (2 marks)
- (v) Compute the probability for $P(d \mid a, \neg b, c)$. (2 marks)
- [Total : 25 marks]

Q2. Figure 2.1 shows a search problem with start state **A** and goal state **E**. The transition cost from one state to the other state is indicated on the respective edge, while the predicted heuristic values (h) for each state to the goal state are given by the table.

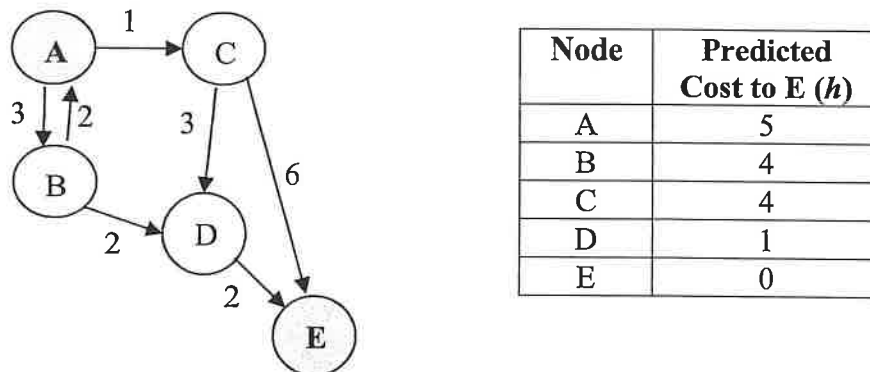


Figure 2.1. A search problem.

- (a) If **Uniform Cost Search (UCS)** is used to find the optimal path to go from state **A** to **E**, draw the **search tree** generated by the UCS algorithm. Show the cost of each node in your drawing. If there are several possible nodes to explore, select the node based on *alphabetical* order. (9 marks)
- (b) If **A* Search** algorithm is used to find the path to go from state **A** to **E**,
- Draw the **search tree** generated by the A* algorithm. Show the **f -value** beside each node in your drawing. (7 marks)
 - Justify whether the heuristic function is **admissible**. (2 marks)
 - Explain why A* Search is more efficient than UCS. (3 marks)
- (c) If a **Depth-First Search (DFS)** tree with alphabetical order is used to find the path to go from **A** to **E**, is the solution complete? Justify your answer. (4 marks)

[Total : 25 marks]

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- Q3. (a) Table 3.1 shows the predicted scores of a binary classifier versus the actual labels for ten test samples. The threshold value of the classifier is set at 0.5.

Table 3.1: Predicted scores of a binary classifier versus actual labels.

Sample #	Actual Label	Predicted Score
1	1	0.78
2	0	0.55
3	1	0.42
4	1	0.86
5	1	0.92
6	1	0.47
7	0	0.02
8	0	0.34
9	1	0.89
10	1	0.71

Determine the following performance measures for the classifier in Table 3.1. Include your calculation in the answer.

- (i) Accuracy (2 marks)
 - (ii) Precision (2 marks)
 - (iii) Recall (2 marks)
 - (iv) F_1 score (2 marks)
- (b) If you want to improve the *recall* rate of the classifier in (a), should you increase or decrease the threshold value of the classifier? Justify your answer. (2 marks)
- (c) Draw a graphical representation of the Markov Decision Process (MDP) and describe the interactions between the key elements in MDP. (7 marks)

UCCD2063 ARTIFICIAL INTELLIGENCE TECHNIQUES**Q3. (Continued)**

- (d) Figure 3.1 shows a simple one-dimensional grid world. There are totally five grids (states) in the grid world. The number in each of the grids indicates the reward received if the agent arrives at the grid. The starting state is at grid #2. The terminal state is at grid #5 with a +2 reward. The agent can only move either left (L) or right (R). You are asked to solve this grid world problem using Q-learning algorithm with learning rate $\alpha=0.5$ and discount factor $\gamma=0.9$.

1	2	3	4	5
0	0	0	0	+2

Figure 3.1. One-dimensional grid world.

- (i) Construct the Q-table for the grid world problem. Initialize the Q-table to zeros. (2 marks)
- (ii) In the first training episode, the agent takes [R, L, R, R, R] action sequence to reach the terminal state. Update the Q-table after the first episode. Show your calculation steps. (2 marks)
- (iii) In the second training episode, the agent takes [R, R, R] action sequence to reach the terminal state. Update the Q-table after the second episode. Show your calculation steps. (4 marks)

[Total : 25 marks]

UCCD2063 ARTIFICIAL INTELLIGENCE TECHNIQUES**Section B (Choose Any ONE (1) Question)**

Q4. (a) Describe the differences between **Discriminative Models** and **Generative Models**. (4 marks)

(b) Table 4.1 shows the input x values and output y values of five training samples.

Table 4.1

x	y
2	25
4	45
6	55
9	55
11	45

Suppose that a **linear** model is used to predict output y from the input x .

- (i) Explain whether this a *regression* task or a *classification* task? (2 marks)
- (ii) Given a linear model ($\theta_0 = 30, \theta_1 = 2$), compute the **RMSE** of the model on the training data. (5 marks)
- (iii) Based on the training data in Table 4.1, discuss whether the linear model overfit, underfit, or just right for the data. (3 marks)
- (iv) Given a second-degree polynomial model ($\theta_0 = -2, \theta_1 = 16, \theta_2 = -1.06$), compute the **RMSE** of the model on the training data and compare it with the result in (ii). (7 marks)
- (v) Polynomial terms can grow very big, suggest a preprocessing method to mitigate the problem. (2 marks)
- (vi) Suggest an approach to control polynomial models from overfitting the training data. (2 marks)

[Total : 25 marks]

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Q5. (a) Table 5.1 shows the training data for classifying output y based on input x .

Table 5.1

x	y
3	no
5	no
7	yes
9	yes

Suppose **logistic regression** is used to perform the classification.

- (i) Write down the hypothesis function $h_{\theta}(x)$ using parameters (θ_0, θ_1) for the model. (2 marks)
- (ii) Given *model A* ($\theta_0 = -3, \theta_1 = 0.5$) and *model B* ($\theta_0 = -1, \theta_1 = 0.3$), determine which model is a better fit for the training data in Table 5.1. Justify your answer by computing their respective classification accuracies. (11 marks)

(b) Given the following cost equation which we wish to minimize:

$$f(x_1, x_2) = 3x_1^2 + 2x_1x_2 + x_2^2 + 3$$

- (i) Derive the update equations for x_1 and x_2 using the *gradient descent* formula below, where η denotes the learning rate. (4 marks)

$$x_j := x_j - \eta \frac{\partial}{\partial x_j} f(x_1, x_2) \text{ for } j = 1, 2$$

- (ii) Given initial values $x_1 = 1, x_2 = 1$ and learning rate $\eta = 0.1$, compute the cost function f for the first TWO (2) iterations using *gradient descent* algorithm. (6 marks)
- (iii) Verify if gradient descent is behaving properly based on cost function calculated in (ii). (2 marks)

[Total : 25 marks]

UCCD2063 ARTIFICIAL INTELLIGENCE TECHNIQUESAppendix

$$Accuracy = \frac{TN + TP}{TN + FP + FN + TP}$$

$$Precision = \frac{TP}{FP + TP}$$

$$F_1 = \frac{TP}{TP + \frac{FN + FP}{2}}$$

$$Recall = \frac{TP}{FN + TP}$$

$$h(\mathbf{x}) = \theta^T \mathbf{x}$$

$$\mathbf{h} = \mathbf{X} \cdot \theta$$

$$MSE(\theta) = \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

$$RMSE(\theta) = \sqrt{MSE(\theta)}$$

$$\nabla_{\theta} MSE(\theta) = \frac{2}{m} \mathbf{X}^T \cdot (\mathbf{X} \cdot \theta - \mathbf{y})$$

$$h_{\theta}(\mathbf{x}) = \sigma(\theta^T \mathbf{x}) = \frac{1}{1 + e^{-\theta^T \mathbf{x}}}$$

$$\nabla_{\theta} MSE(\theta) = \frac{1}{m} \mathbf{X}^T \cdot (h_{\theta}(\mathbf{x}) - \mathbf{y})$$

$$P(A, B) = P(A|B)P(B)$$

$$P(A, B) = P(B|A)P(A)$$

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

$$P(A) + P(\neg A) = 1$$

$$Q(s, a) = Q(s, a) + \alpha(R(s, a) + \gamma \max_{a'} Q(s', a') - Q(s, a))$$